

For this project, several electronic components were integrated to create a functional and automated parking management system. These include an LCD display to show the number of available parking spaces, an ultrasonic sensor to detect incoming vehicles, a servo motor that operates as the entrance gate, a buzzer for audio indication, and Light Dependent Resistors (LDRs) installed at each parking spot to determine whether a space is occupied or available. Each parking spot is also equipped with an LED indicator to visually display its status.

Let us now take a look at how the system works.

As a vehicle approaches the parking lot entrance, the driver can immediately see the number of available parking spaces displayed on the LCD screen. The ultrasonic sensor continuously monitors the entrance area by sending out ultrasonic waves till a specified distance, which when reflected back detects the presence of an approaching vehicle.

If at least one parking space is available, the system activates the servo motor, opening the gate automatically. At the same time, the buzzer sounds to indicate that the gate is in operation. Once the vehicle enters the parking lot, the gate closes automatically behind it.

Inside the parking lot, available spaces are clearly identified by green LEDs. The driver can choose any free parking spot and park the vehicle.

When a vehicle occupies a parking space, the LDR sensor detects the change in light intensity caused by the vehicle overhead. The system then marks that space as occupied, switches off the corresponding green LED, and updates the LCD display in real time to reflect the reduced number of available parking spaces.

If the parking lot becomes completely full, the system prevents additional vehicles from entering. In this case, the LCD screen displays the message: "Sorry, the lot is full," and neither the gate nor the buzzer will activate.